🎧

# TwinMind Automatic Speech Recognition (ASR) API

Welcome to the **TwinMind Automatic Speech Recognition (ASR) API** powered by our Ear-3 model. With industry-leading 5.26% Word Error Rate and 3.8% Diarization Error Rate, our API delivers unmatched accuracy for speech-to-text conversion.

This API enables developers to transcribe audio files into text with exceptional accuracy and speaker identification. Built for scalability and easy integration, the ASR API supports 140+ languages at just $0.2/hour, making it both the most comprehensive and cost-effective solution available. Process files up to 24 hours long with speeds 18× faster than real-time while maintaining enterprise-grade security with HIPAA compliance.

## 🎯 Choose Your Model

TwinMind offers two ASR models optimized for different use cases. Select the model that best balances your needs for accuracy, speed, and cost:

| Feature | Ear-3-Pro | Ear-3-Fast |
|---|---|---|
| **Transcription Accuracy** | Highest accuracy (5.26% WER) | High accuracy (slightly lower than Pro) |
| **Speaker Diarization** | Most accurate speaker identification (3.8% DER) | Good speaker identification (higher DER than Pro) |
| **Translation Quality** | Superior translation accuracy | Good translation quality |
| **Processing Speed** | 18× faster than real-time | 25× faster than real-time |
| **Cost** | $0.20 per hour | $0.12 per hour |
| **Best For** | Critical applications, medical/legal transcription, complex multi-speaker scenarios | High-volume processing, quick turnaround needs, cost-sensitive applications |

## 🔐 Authentication

All requests to user endpoints must include a valid **API key** in the HTTP Authorization header.

### Header Format

```
Authorization: Bearer sk_live_yourapikey
```

### Error Responses

If the key is missing or invalid, the API will respond with:

```
{
  "error": "Unauthorized",
  "message": "Invalid or missing API key"
}
```

> **Security Note:** Keep your API key confidential — it provides full access to your account's usage and balance.

# 1. Transcribe Audio API

**Endpoint:** `POST https://api.twinmind.dev/v1/transcribe`

**Purpose:** Convert audio files to text with support for speaker identification (diarization), contextual hints, and multilingual transcription.

## 📋 Request Parameters

| Parameter | Type | Required | Description |
|---|---|---|---|
| **file** | File | ✅ | Audio file to transcribe.<br>Formats: `mp3` , `m4a` , `wav` , `flac` , `ogg` , `aac` . |
| **model** | String | ❌ | Model to use for transcription. Options: `ear-3-pro` (default) or `ear-3-fast` . |
| **language** | String | ❌ | Audio language. Default: `auto detect` . Also supports: `spanish` , `french` , `german` , `italian` , `portuguese` . |
| **num_speakers** | Integer | ❌ | Number of distinct speakers (1–10). Enables speaker labeling. |
| **prompt** | String | ❌ | Contextual hints about domain-specific vocabulary or terminology. |
| **translate** | String | ❌ | If `true` , translates transcription to the specified language. If language not provided then defaults to English translation. |

## 🧾 Request Example

```
curl -X POST https://api.twinmind.dev/v1/transcribe \
  -H "Authorization: Bearer sk_live_xxxxx" \
  -F "file=@/path/to/audio.m4a" \
  -F "language=english" \
  -F "num_speakers=2" \
  -F "prompt=Meeting about quarterly revenue projections, TwinMind strategic initiatives, AI model optimization, NLP infrastructure, speech recognition accuracy metrics, Word Error Rate benchmarks"
```

## ⚙️ Processing & Billing

- Transcription occurs in real-time (synchronous processing)
- Billing is based on **audio duration** only: **$0.20 per hour** for **Ear-3-Pro** and **$0.12 per hour** for **Ear-3-Fast**

## ✅ Successful Response

```
{
  "success": true,
  "status": "completed",
  "transcription": {
    "0": {
      "timestamp": "00:00:00 - 00:00:15",
      "speaker": "Speaker A",
      "text": "Hello, welcome to today's meeting.",
      "start_seconds": 0,
      "end_seconds": 15
    },
    "1": {
      "timestamp": "00:00:15 - 00:00:30",
      "speaker": "Speaker B",
      "text": "Thank you, let's begin with the Q3 targets.",
      "start_seconds": 15,
      "end_seconds": 30
    }
  }
}
```

---

## ⚠️ Error Handling

| Code | Meaning | Example |
|---|---|---|
| **400** | Invalid request | `{"detail": "Missing audio file"}` |
| **401** | Unauthorized | `{"error": "Unauthorized"}` |
| **402** | Insufficient credits | `{"detail": "Insufficient credits: need $4.50, have $2.30"}` |
| **415** | Unsupported file type | `{"detail": "Unsupported audio format"}` |
| **429** | Rate limit exceeded | `{"detail": "Rate limit exceeded: retry after 60s"}` |
| **500** | Internal server error | `{"detail": "Internal Server Error"}` |

# 2. Asynchronous Transcription

For large files or batch processing, use our asynchronous endpoints. Submit jobs, check status, and retrieve results separately—ideal for production systems processing multiple files.

## 2A. Submit Transcription Job

**Endpoint:** `POST https://api.twinmind.dev/v1/transcribe-async`

**Purpose:** Submit audio files for asynchronous transcription processing. Returns immediately with a job ID for later status checking and result retrieval.

### 📋 Request Parameters

| Parameter | Type | Required | Description |
|---|---|---|---|
| **file** | File | ✅ | Audio file to transcribe.<br>Formats: `mp3` , `m4a` , `wav` , `flac` , `ogg` , `aac` . |
| **model** | String | ❌ | Model to use for transcription. Options: `ear-3-pro` (default) or `ear-3-fast` . |
| **language** | String | ❌ | Audio language. Default: `auto detect` . Also supports: `spanish` , `french` , `german` , `italian` , `portuguese` . |
| **num_speakers** | Integer | ❌ | Number of distinct speakers (1–10). |
| **prompt** | String | ❌ | Contextual hints about domain-specific vocabulary or terminology |
| **translate** | String | ❌ | If `true` , translates transcription to the specified language. If langauge not provided then defaults to English translation. |

### 🧾 Request Example

```
curl -X POST https://api.twinmind.dev/v1/transcribe-async \
  -H "Authorization: Bearer sk_live_xxxxx" \
  -F "file=@/path/to/audio.m4a" \
  -F "language=english" \
```

```
  -F "num_speakers=2" \
  -F "prompt=Technical terms: AI, ML, API, cloud computing"
```

### ✅ Successful Response

```
{
  "success": true,
  "status": "waiting",
  "transcription_job_id": "550e8400-e29b-41d4-a716-446655440000"
}
```

### ⚙️ Job Status Flow

| Status | Description |
| --- | --- |
| **waiting** | Job queued, waiting to be processed |
| **processing** | Transcription in progress |
| **completed** | Transcription finished, result available |
| **failed** | Transcription failed (credits refunded) |

## 2B. Check Job Status

**Endpoint:** `GET https://api.twinmind.dev/v1/transcribe-async/status/{job_id}`

**Purpose:** Check the current status of an asynchronous transcription job.

### 📋 Path Parameters

| Parameter | Type | Required | Description |
| --- | --- | --- | --- |
| **job_id** | UUID | ✅ | Job ID returned from the transcribe-async endpoint |

### 🧾 Request Example

```
curl -X GET https://api.twinmind.dev/v1/transcribe-async/status/550e8400-e29b-41d4-a71
6-446655440000 \
  -H "Authorization: Bearer sk_live_xxxxx"
```

### ✅ Response Examples

**Job in Progress:**

```
{
  "success": true,
```

```
  "job_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "processing"
}
```

**Job Completed:**

```
{
  "success": true,
  "job_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "completed"
}
```

**Job Failed:**

```
{
  "success": true,
  "job_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "failed",
  "error_message": "Audio file corrupted or unsupported format"
}
```

### ⚠️ Error Responses

| Code | Reason | Response |
|---|---|---|
| **404** | Job not found | `{"detail": "Job not found"}` |
| **401** | Unauthorized | `{"error": "Unauthorized"}` |

## 2C. Retrieve Result

**Endpoint:** `GET https://api.twinmind.dev/v1/transcribe-async/result/{job_id}`

**Purpose:** Retrieve the completed transcription result for a finished job.

### 📋 Path Parameters

| Parameter | Type | Required | Description |
|---|---|---|---|
| **job_id** | UUID | ✅ | Job ID returned from the transcribe-async endpoint |

### 🧾 Request Example

```
curl -X GET https://api.twinmind.dev/v1/transcribe-async/result/550e8400-e29b-41d4-a716
-446655440000 \
```

```
-H "Authorization: Bearer sk_live_xxxxx"
```

## ✅ Successful Response


```json
{
  "success": true,
  "job_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "completed",
  "transcription": {
    "0": {
      "timestamp": "00:00:00 - 00:00:15",
      "speaker": "Speaker A",
      "text": "Hello, welcome to today's meeting.",
      "start_seconds": 0,
      "end_seconds": 15
    },
    "1": {
      "timestamp": "00:00:15 - 00:00:30",
      "speaker": "Speaker B",
      "text": "Thank you, let's begin with the Q3 targets.",
      "start_seconds": 15,
      "end_seconds": 30
    },
    "metadata": {
      "total_segments": 2,
      "duration": "00:05:30",
      "total_duration_seconds": 330,
      "vocab": ["AI", "ML", "API"]
    }
  }
}
```


## ⚠️ Error Responses

| Code | Reason | Response |
|---|---|---|
| **400** | Job not completed | `{"detail": "Job not completed yet. Current status: processing"}` |
| **404** | Job not found | `{"detail": "Job not found"}` |
| **404** | Result unavailable | `{"detail": "Transcription not available"}` |
| **401** | Unauthorized | `{"error": "Unauthorized"}` |

## 💡 Sync vs Async: When to Use Which?

| Feature | Sync ( `/v1/transcribe` ) | Async ( `/v1/transcribe-async` ) |
|---|---|---|
| **Response Time** | Immediate (waits for completion) | Immediate (returns job ID) |
| **Best For** | Short files (<5 min) | Long files, batch processing |
| **Connection** | Must stay open | Fire and forget |
| **Result Retrieval** | In same response | Separate endpoint |
| **Timeout Risk** | Yes (for long files) | No |
| **Retry Logic** | Manual | Automatic |

# 3. Retrieve Usage History

**Endpoint:** `GET https://api.twinmind.dev/v1/usage`

**Purpose:** Access detailed usage statistics including transaction history, credit consumption, and current account balance.

## 📋 Query Parameters

| Parameter | Type | Required | Default | Max | Description |
|---|---|---|---|---|---|
| **limit** | Integer | ❌ | 50 | 500 | Number of transactions to return |

## 🧾 Request Example

```
curl -X GET "https://api.twinmind.dev/v1/usage?limit=20" \
  -H "Authorization: Bearer sk_live_xxxxx"
```

## ✅ Successful Response Example

```
{
  "current_balance": 45.50,
  "transactions": [
    {
      "transaction_id": "tx-987...",
      "amount": -4.50,
      "transaction_type": "usage",
      "description": "Transcription - 120.5s audio",
      "job_id": "job-abc...",
```

```
      "created_at": "2025-10-06T22:30:00Z"
    },
    {
      "transaction_id": "tx-654...",
      "amount": 50.00,
      "transaction_type": "welcome_bonus",
      "description": "Welcome bonus - $50.0",
      "created_at": "2025-10-06T21:53:53Z"
    }
  ]
}
```

## 4. Get Current Balance

**Endpoint:** `GET https://api.twinmind.dev/v1/usage/balance`

**Purpose:** Retrieves the user's **current credit balance** only — a lightweight endpoint for frequent balance checks.

### 📋 Request

```
curl -X GET https://api.twinmind.dev/v1/usage/balance \
  -H "Authorization: Bearer sk_live_xxxxx"
```

### ✅ Response

```
{
  "current_balance": 45.50
}
```

## ⚙️ Rate Limits

To ensure system stability and fair usage across all users, the following rate limits apply to all API keys:

| Limit Type | Default | Description |
|---|---|---|
| **RPM** | 60 | Requests per minute |
| **RPD** | 1000 | Requests per day |

**When limit is exceeded:**

```
{
  "detail": "Rate limit exceeded: RPM, retry after 60s"
}
```

## ⚙️ Processing & Billing

- Credits are reserved when job is submitted
- Billing is based on **audio duration** only:
  - **Ear-3-Pro:** $0.2 per hour
  - **Ear-3-Fast:** $0.12 per hour
- For production systems processing multiple files or files longer than 5 minutes, use async endpoints to avoid timeout issues and enable better resource management

## 🚨 Error Reference Guide

| Code | Type | Description |
| --- | --- | --- |
| 400 | Bad Request | Invalid parameters, malformed request body, or missing required fields |
| 401 | Unauthorized | Invalid API key, expired token, or insufficient permissions |
| 402 | Payment Required | Insufficient account credits to process the requested operation |
| 404 | Not Found | Requested resource, user account, or endpoint does not exist |
| 415 | Unsupported Media Type | Audio file format not supported or corrupted media file |
| 429 | Too Many Requests | API rate limit exceeded; implement exponential backoff strategy |
| 500 | Internal Server Error | Server-side issue; retry after a short delay or contact support |

## 🧭 Quick Integration Overview

| Task | Endpoint | Method | Description |
| --- | --- | --- | --- |
| Upload and transcribe audio | `https://api.twinmind.dev/v1/transcribe` | POST | Transcribes audio with optional context and diarization |
| Upload and transcribe audio (async) | `https://api.twinmind.dev/v1/transcribe-async` | POST | Submit transcription job and receive job ID immediately |
| Check async job status | `https://api.twinmind.dev/v1/transcribe-async/status/{job_id}` | GET | Check the current status of an async |

| | | | |
|---|---|---|---|
| | | | transcription job |
| Retrieve async transcription result | `https://api.twinmind.dev/v1/transcribe-async/result/{job_id}` | GET | Fetch completed transcription for finished async job |
| Retrieve usage history | `https://api.twinmind.dev/v1/usage` | GET | Returns transactions and current balance |
| Fetch balance only | `https://api.twinmind.dev/v1/usage/balance` | GET | Lightweight check for available credits |

### 📞 Support

If you encounter issues, please reach out to our team:

**Email:** support@twinmind.com